



Green University of Bangladesh

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EHealth- Connecting A Nations Health

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Chapter 1

Introduction

1.1 Overview

The healthcare system in Bangladesh still faces major challenges due to the lack of a unified digital health record platform. Medical reports and prescriptions are often stored in paper form or isolated systems, leading to frequent loss of records, repeated diagnostic tests, delayed treatment, and limited access to a patient's complete medical history. Doctors are unable to view previous diseases, medications, and test results, increasing the risk of incorrect treatment. In addition, the absence of proper doctor verification and prescription tracking makes it difficult to identify fake doctors and unauthorized prescriptions. Rural patients also face barriers in sharing medical records with urban specialists, while the government lacks real-time data for effective healthcare planning.[1] EHealth is proposed as a centralized digital health record system to address these challenges. The platform will store all patient medical records in one secure location, allow instant access to complete treatment histories, reduce duplicate tests, support telemedicine, and improve doctor and hospital accountability. HealthSetu will also enable real-time disease monitoring and data-driven health policies, helping Bangladesh move toward a more efficient, transparent, and digital healthcare system.

1.2 Problem Definition

The healthcare sector in Bangladesh operates through a highly fragmented medical record management process. Patient medical information such as diagnostic reports, prescriptions, and treatment histories, is maintained across multiple hospitals, clinics, and diagnostic centers without a standardized or centralized system. In many cases, records are stored in paper-based or partially digital formats, which makes long-term preservation, retrieval, and sharing of information unreliable and inefficient.[2][4] This fragmentation creates several critical issues, including loss of medical records, repeated diagnostic tests, delayed treatment, and increased risk of medical errors due to incomplete patient history. The absence of a unified platform also limits effective verification of doctors and prescriptions, allowing unauthorized practices to persist. Furthermore, healthcare authorities lack real-time visibility into disease

patterns, hospital compliance, and resource utilization, making data-driven decision-making difficult. These challenges highlight the need for a centralized, secure, and transparent digital health record system tailored to the healthcare environment of Bangladesh.

1.3 Motivation

The motivations behind this project are :

- Several developed countries, such as Finland and Singapore, have successfully implemented centralized digital health record systems, demonstrating improved treatment accuracy, transparency, and efficiency in healthcare delivery.
- In Bangladesh, medical records remain fragmented and often paper-based, leading to loss of reports, repeated diagnostic tests, and delayed or incorrect treatment.[3]
- Doctors frequently lack access to complete patient medical histories, which increases clinical risk and reduces the quality of healthcare services.
- The absence of effective mechanisms for doctor verification and prescription tracking allows fake doctors and unauthorized medical practices to continue.
- Rural patients face difficulties in sharing medical records with urban specialists, limiting the effectiveness of telemedicine and remote healthcare services.
- Government health authorities lack real-time, reliable healthcare data for disease monitoring and evidence-based policy planning, motivating the need for a unified digital health record system like HealthSetu.

1.4 Objectives

The objectives of this project are:

- To design a centralized digital health record management system that securely stores patients' medical reports, prescriptions, and treatment histories in one unified platform.
- To enable doctors and healthcare institutions to access complete and accurate patient medical histories, thereby reducing medical errors and unnecessary duplicate diagnostic tests.
- To establish a structured mechanism for doctor identification and prescription tracking to improve accountability and reduce unauthorized medical practices.

- To support efficient sharing of medical records between patients and healthcare providers, including remote and telemedicine-based consultations.
- To propose a scalable and future-ready system architecture that can incorporate advanced technologies such as artificial intelligence, blockchain, and enhanced security mechanisms in later phases[5][6]
- To assist government health authorities in monitoring disease trends and health-care service performance using aggregated and anonymized data.

Chapter 2

Literature Review

Several charity platforms and organizations provide medical assistance to patients. However, most existing systems lack strict patient verification, detailed cost validation, and transparency in fund usage. Research shows that digital health platforms with proper verification mechanisms can significantly reduce fraud and increase donor trust. These studies highlight the importance of a secure and transparent charity management system like Lifeline Charity Platform.

Existing System	Services Provide	Limitations vs Our eHealth Platform
P1: Traditional paper-based health record systems	Manual storage of prescriptions, medical reports, and patient history within individual hospitals or clinics	Prone to data loss and duplication; lacks remote access, data sharing, and long-term preservation of records[7]
P2: Hospital Management Systems (HMS)	Digital management of patient records, ap-pointments, billing, and diagnostic services within a single hospital	Data is siloed and restricted to one institution; no interoperability or nationwide access[8]
P3: Standalone Electronic Health Record (EHR) systems	Electronic storage and retrieval of patient medical history and clinical data	Limited interoperability; restricted patient access and minimal user control over personal data[9]
P4: Cloud-based health record platforms	Online storage, access, and basic sharing of medical data	Security and privacy concerns; lack of standardized access control and government-level governance
P5: Health Information Exchange (HIE) systems	Data exchange between multiple healthcare institutions	Complex implementation; limited patient involvement and unsuitable for unified personal health record management[10]

Chapter 3

Methodology

3.1 Prototype Design

The prototype design of the E-Health system focuses on simplicity, security, and ease of navigation to ensure a positive user experience for both patients and healthcare providers. The interface is designed with a clean layout, clear typography, and minimal steps to access critical features. Patients can log in to view their health records, prescriptions, test reports, and appointment history through a centralized dashboard. Verified doctors, hospitals, and diagnostic centers have a separate role-based interface that allows secure uploading and management of medical records. The overall UI/UX emphasizes intuitive navigation, fast access to information, and visual clarity while maintaining strict privacy and data security standards.

3.2 Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) describes the functional and non-functional requirements of the Lifeline Charity Platform.

1. Functional Requirements

- User Authentication
- Role-Based Access
- Patient Management
- Doctor Management
- Diagnostic Management
- Hospital Management
- Medical Record Management
- Prescription Management
- Diagnostic Report Management
- Secure Data Access

2. Non-Functional Requirements

- Security and Privacy
- Usability and Reliability
- Performance and Maintainability
- Scalability

3.3 SDLC

Selecting an appropriate Software Development Life Cycle (SDLC) model is a crucial step for the successful development of a software project. The SDLC model defines the structured approach used to plan, design, develop, test, deploy, and maintain the system. A suitable model ensures better project management, improved quality, and timely completion of the project.[12]

Priority	Criteria	SDLC Models					
		Waterfall	V-Shape	Iterative	Spiral	Agile	Prototype
5	Well-Known Requirement	Yes	Yes	No	Yes	No	No
3	Technological Knowledge	Yes	Yes	No	No	No	No
6	Efficiency	No	Yes	Yes	Yes	No	No
3	Risk Analysis	No	No	No	Yes	No	No
5	User Test ability	No	No	Yes	Yes	Yes	No
5	Dependability & Security	No	Yes	Yes	Yes	No	No
3	Time Consuming	Yes	No	No	Yes	No	No
Total (30)		11	19	16	27	5	0

Table 3.1: SDLC model Selection

The Spiral model was selected for the EHealth project because it supports continuous risk analysis, which is critical when handling sensitive medical data and system security. Additionally, its iterative development and regular user feedback ensure higher reliability, accuracy, and compliance with evolving healthcare requirements.

3.4 Data Flow Diagram

A Data Flow Diagram (DFD) is used in this project to visually represent how data

flows through the system and how different processes interact with external entities and data stores. It helps in understanding the system's functional behavior by clearly illustrating input, processing, storage, and output of data.

DFD improves system clarity and communication by providing a structured view of data movement, making the design easier to analyze, validate, and implement accurately

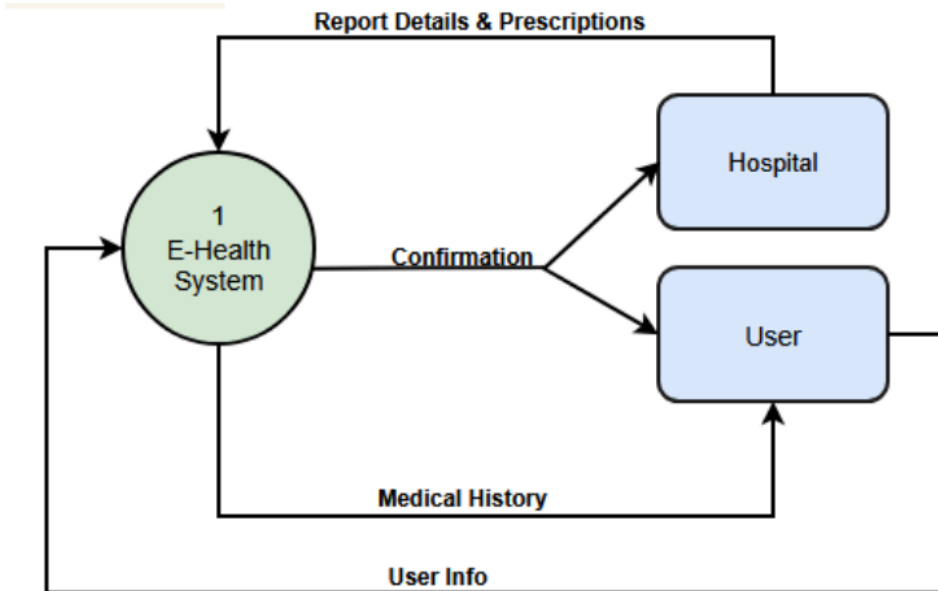


Figure 3.1: Level 0 DFD of E-Health

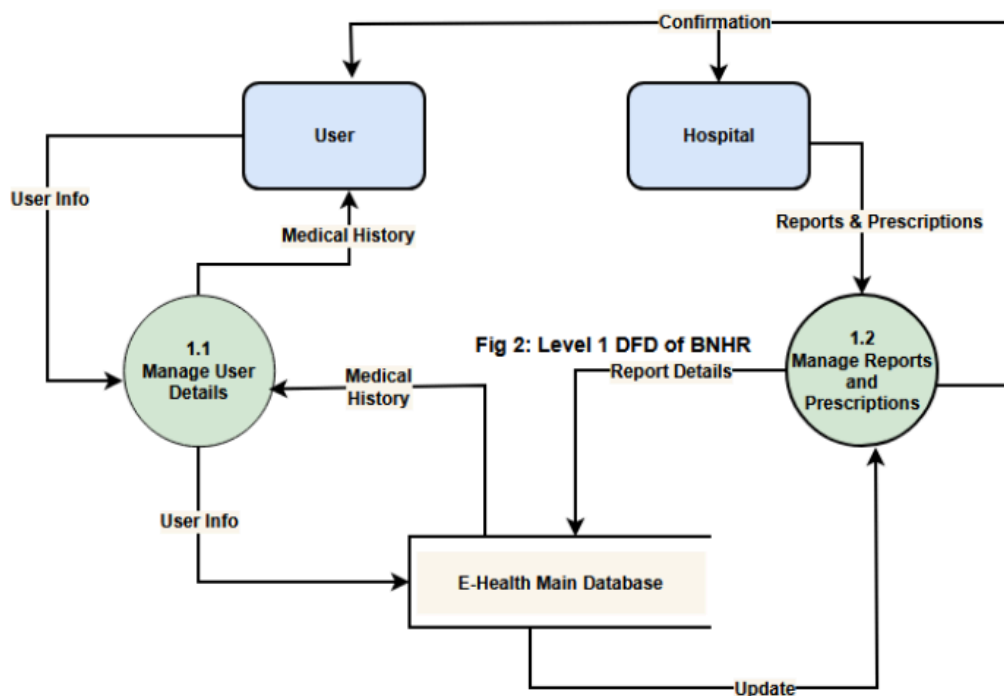


Figure 3.2: Level 1 DFD of E-Health

3.5 UML Use Case Diagram

A UML Use Case Diagram is used in this project to represent the functional requirements of the system from the user's perspective. It illustrates the interactions between different actors (such as users, administrators, or external systems) and the system through various use cases.

The use case diagram helps in clearly defining system boundaries, identifying user roles, and ensuring that all required functionalities are captured, making it an effective tool for requirement analysis and system design.

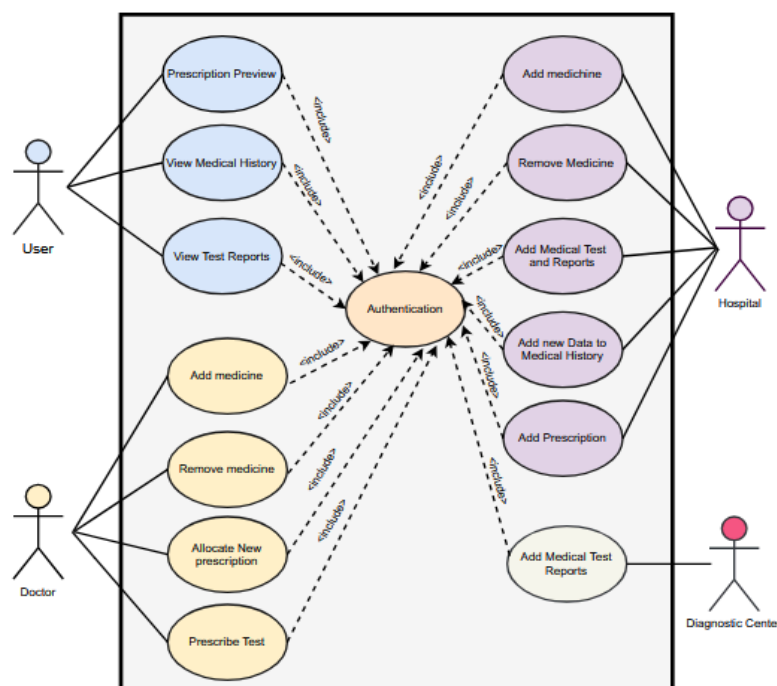


Figure 3.3: UML Use Case Diagram for EHealth

3.6 UML Sequence and Communication Diagram

The UML Sequence and Communication Diagram is used in this project to represent the interaction among system components for a specific process. It shows how objects communicate with each other in a logical sequence while also highlighting their relationships and message flow.

This diagram helps in understanding the system's dynamic behavior and object collaboration, ensuring that the interaction logic is clearly defined and correctly implemented.

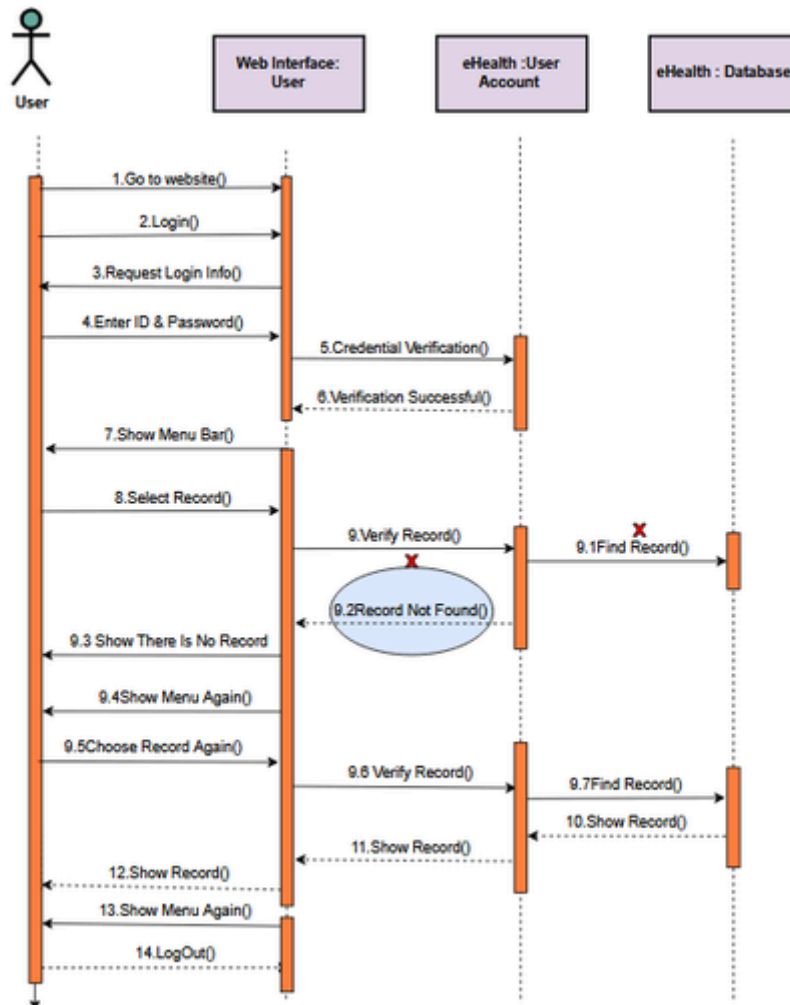


Figure 3.4: UML s&c diagram of a user of E-Health

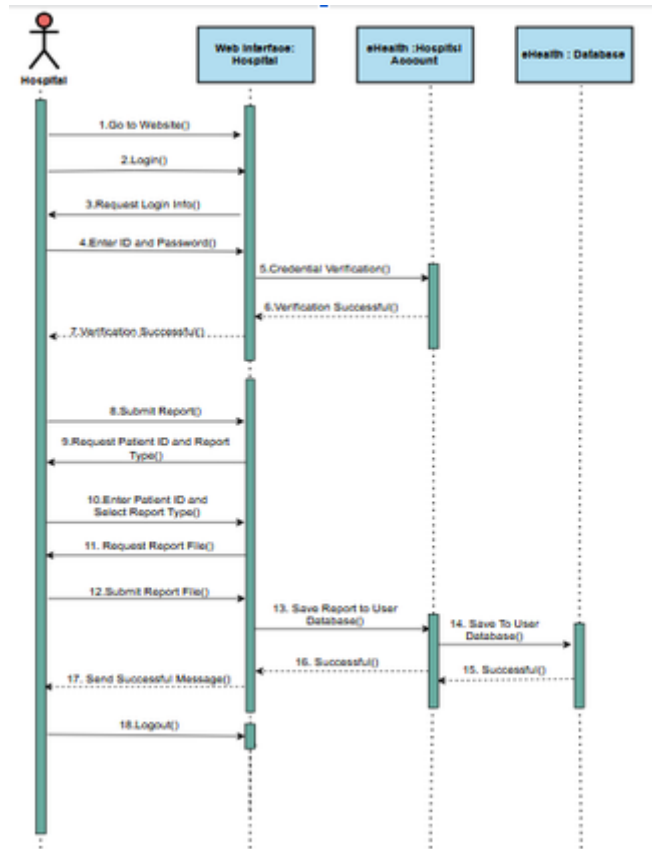


Figure 3.5: UML s&c diagram of a hospital of E-Health

3.7 UML Class Diagram

A UML (Unified Modeling Language) Class Diagram is a type of static structure diagram that shows the classes of a system, their attributes, methods (operations), and the relationships among objects. It is widely used in software engineering to visually represent the design of a system.

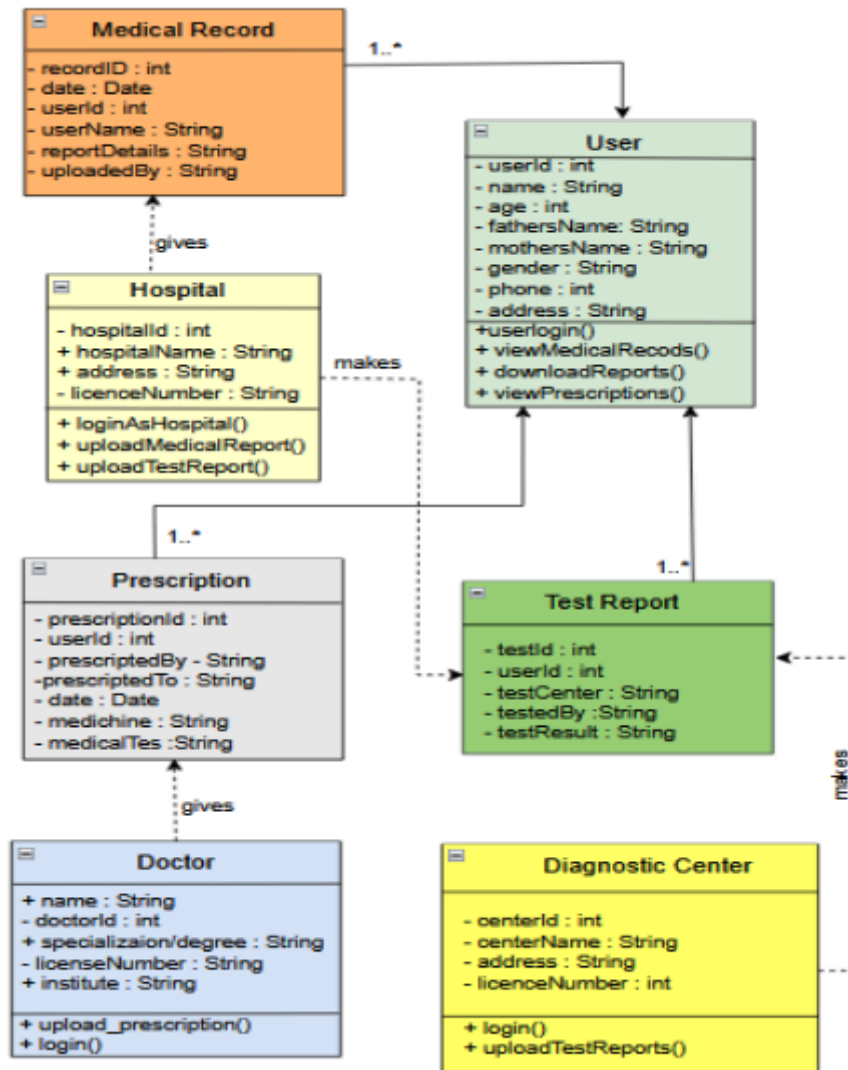


Figure 3.8: UML class diagram for EHealth

3.8 High Level Design

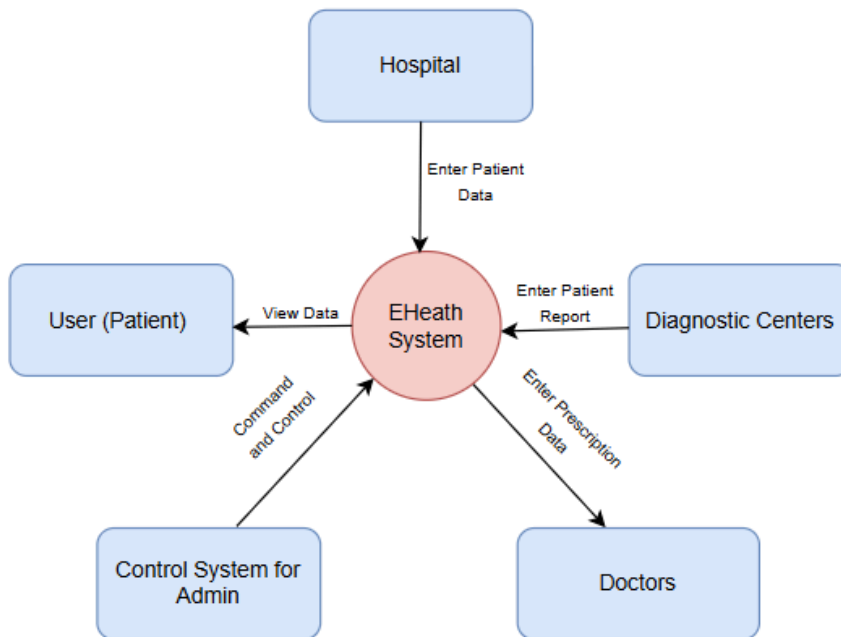


Fig 3.9 : Higher Level Design of EHealth

3.9 Low Level Design

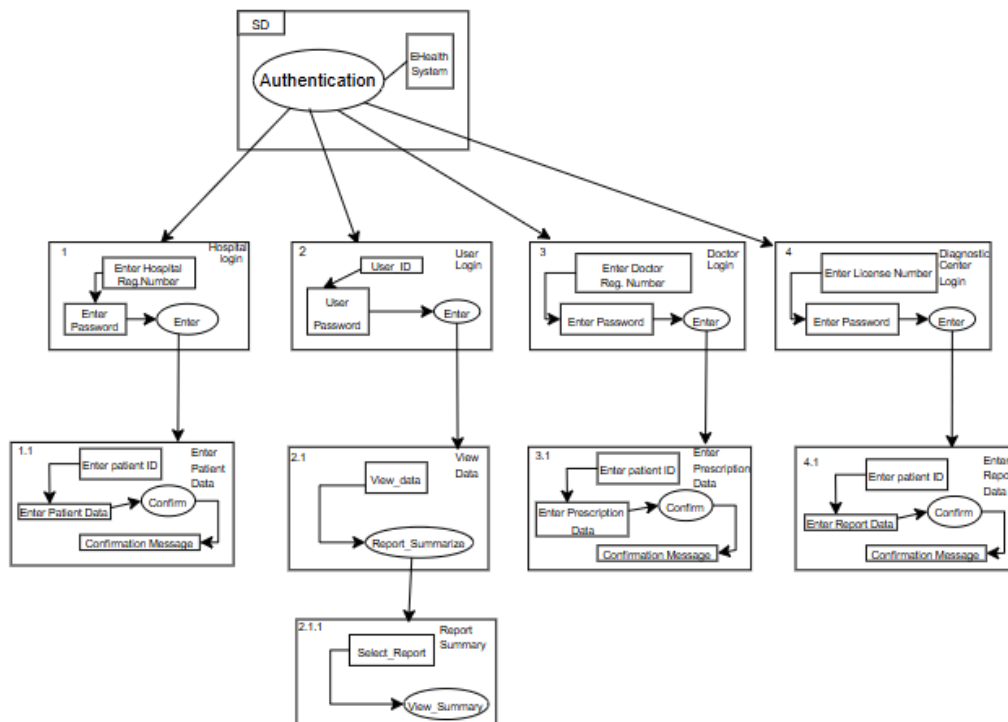


Fig 3.10 : Low Level System Design of EHealth



3.10 ER Diagram

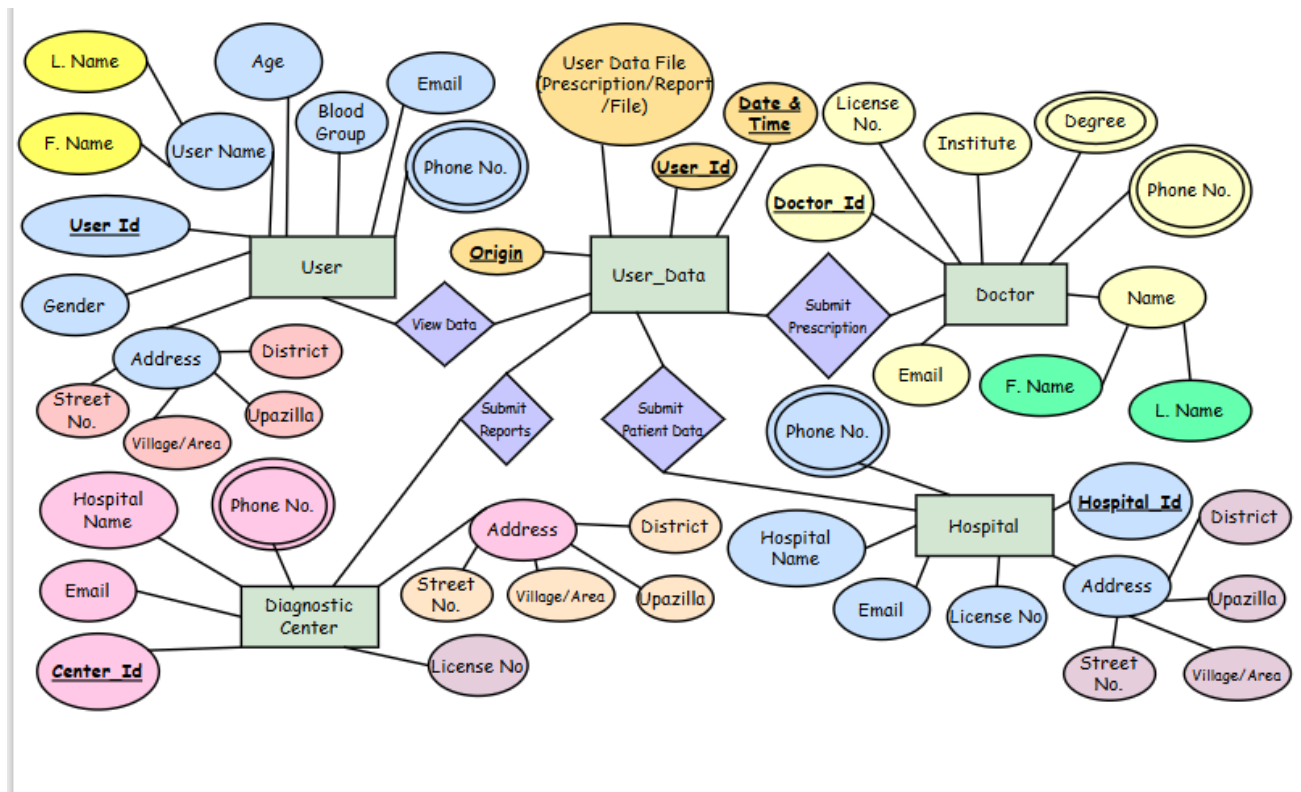


Fig 3.11 Entity Relationship Diagram

Chapter 4

System Architecture & Database Design

4.1 User Interface (UI) Design

1. Home page

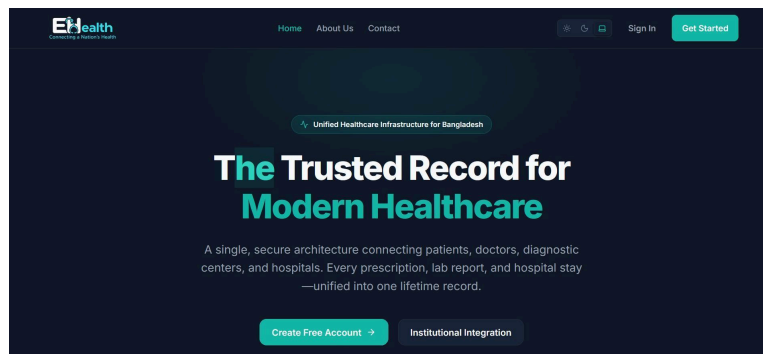


Figure 4.4: Home page of E-Health

2. Users Dashboard

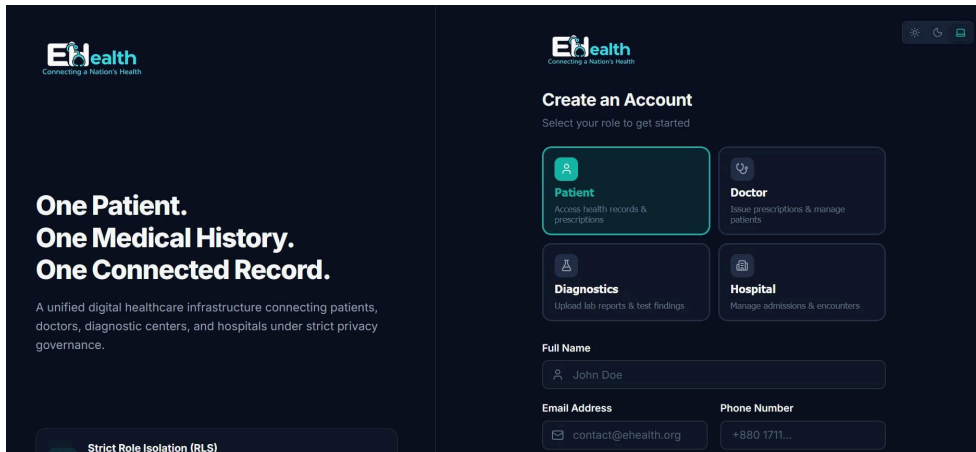


Figure 4.5: Users dashboard

3. Patient Vault

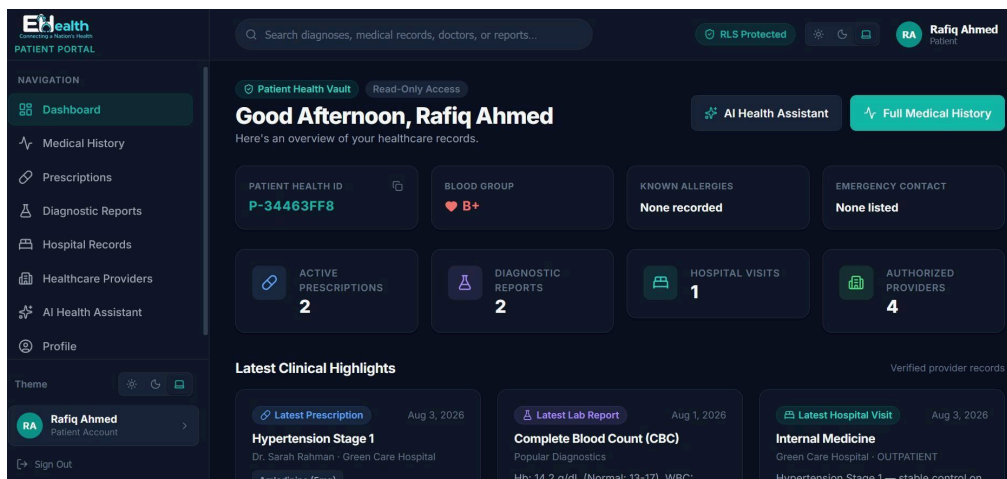


Figure 4.6: Patient data

4. Doctors Dashboard

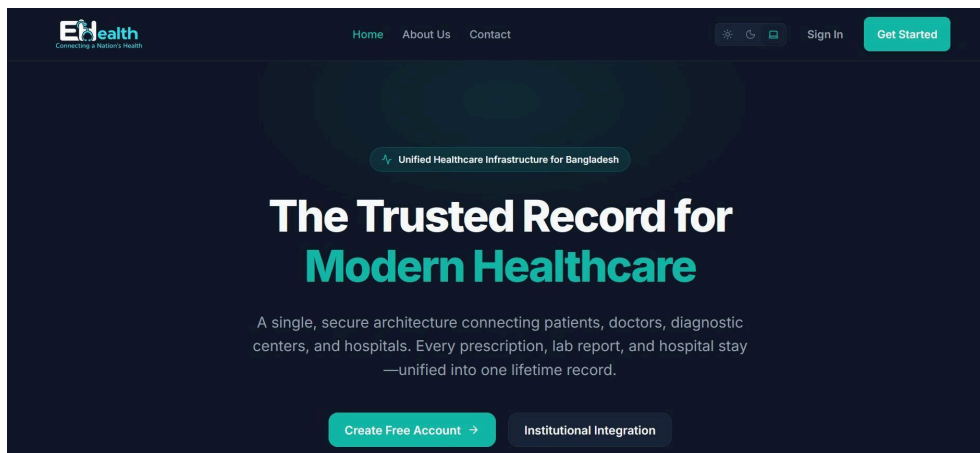


Figure 4.7: Doctor Dashboard

5. Hospital Dashboard

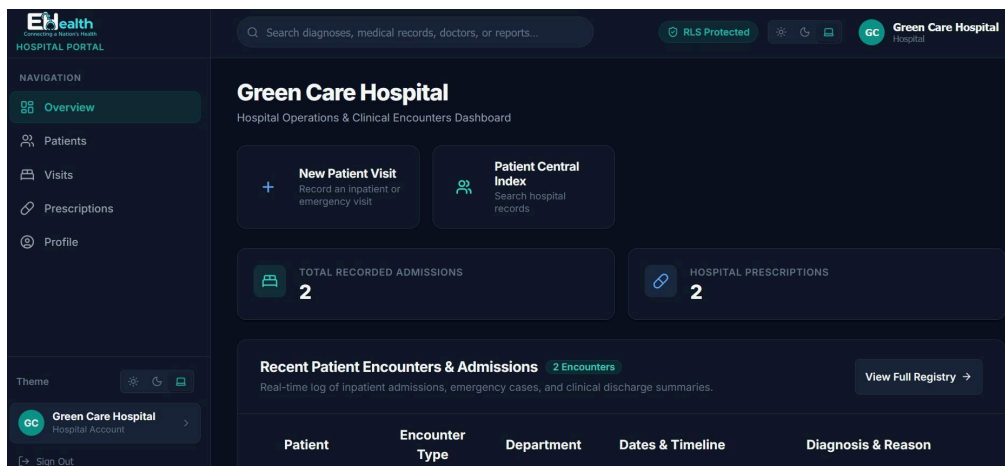


Figure 4.8: Hospital Dashboard

Chapter 5

System Implementation & Testing

5.1 Technology Stack & Environment

5.1.1 Software & Technology Stack

Frontend:

- React 19 — Used to develop the interactive user interface.
- Vite — Used as the development and build tool.
- React Router v7 — Used for application routing and navigation.

Backend & API:

- Supabase — Used for backend services and database integration.
- Supabase Auth — Used for user registration and authentication.

Database:

- PostgreSQL — Used to store and manage patient, doctor, diagnostic, hospital, prescription, and medical record data.

Security:

- Row Level Security (RLS) — Used to control access to sensitive healthcare data based on user authorization.

Storage:

- Supabase Storage — Used for storing medical documents and supporting files securely.

UI & Icons:

- CSS / CSS Variables — Used for styling and interface customization.
- Lucide React — Used for interface icons.

Development Environment:

- Visual Studio Code (VS Code) — Used as the primary code editor.
- Git & GitHub — Used for version control and source-code management.
- Operating System - Windows/Linux
- Platform: Web-based application

5.2 Core Module Implementation

User Authentication:

- User registration and login are implemented using Supabase Authentication.
- The system verifies user credentials before providing access. Authenticated users can access protected system features.

Role-Based Access:

- The system supports Patient, Doctor, Diagnostics, and Hospital roles.
- Each role has a separate portal and specific permissions.
- Unauthorized users are restricted from accessing protected resources.

Patient Module:

- Patients can view their medical history, prescriptions, diagnostic reports, and hospital records.
- Patients can view their authorized healthcare providers.
- Patient information is displayed through a centralized healthcare timeline.

Doctor Module:

- Doctors can search for authorized patients.
- Doctors can view accessible patient medical records and clinical history.
- Doctors can create and manage electronic prescriptions.

Diagnostics Module:

- Diagnostics users can search for patients.
- They can create and manage diagnostic reports.
- Supporting medical documents can be attached to patient records.

Hospital Module:

- Hospitals can manage patient admissions and outpatient visits.
- The module supports emergency and department-related patient information.
- Hospitals can maintain relevant patient healthcare records.

5.3 Test Cases & Execution Results

1. User Login

Module Name: Authentication

Test Case ID: AUTH_01

Tester Name: Sinthia Akter

Test Case Description: To check whether a registered user can successfully login using valid email and password.

Prerequisites:

1. Stable internet connection.
2. A registered user account exists.
3. Browser is available.

Environmental Information:

1. OS: Windows/Linux
2. System: Laptop/Desktop

Test Scenario: Checking that a registered user can successfully log in and access the appropriate role-based dashboard.

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
AUTH_01	1. Open Login	/login	Login page should open.	Login page opened successfully.	Pass	No issue.
AUTH_01	2. Enter Email	Valid email	Email should be accepted.	Email accepted successfully.	Pass	No issue.
AUTH_01	3. Enter Password	Valid password	Password should be accepted.	Password accepted successfully.	Pass	No issue.
AUTH_01	4. Click Sign In	Valid credentials	User should be authenticated and redirected to the correct portal.	User authenticated and redirected to the appropriate role portal.	Pass	Login works properly.

2. Role-Based Access Control

Module Name:Authorization

Test Case ID: AUTH_02

Tester Name: Mashuk

Test Case Description: To check whether users can access only the portal associated with their assigned role.

Prerequisites:

1. Stable internet connection.
2. Valid beneficiary account exists.
3. Browser is available.

Environmental Information:

1. OS: Windows/Linux/Mac
2. System: Laptop/Desktop

Test Scenario: A logged-in user attempts to access role-specific routes.

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
AUTH_02	1. Login as Patient	Valid Patient credentials	Patient should access /patient.	Patient route is protected for Patient role.	Pass	Role restriction implemented.
AUTH_02	2. Attempt Doctor route	/doctor	Patient should not access Doctor portal.	Doctor route allows only Doctor role.	Pass	Access control implemented.
AUTH_02	3. Login as Doctor	Valid Doctor credentials	Doctor should access /doctor.	Doctor route is protected for Doctor role.	Pass	No issue found.

AUTH_02	4. Attempt Hospital route as Doctor	/hospital	Doctor should not access Hospital portal.	Hospital route allows only Hospital role.	Pass	Role isolation imple- mented.
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3. Patient Medical History

Module Name: Medical History

Test Case ID: PAT_01

Tester Name: Imran

Test Scenario: Patient opens the Medical History section and views the unified medical record timeline.

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
PAT_01	1. Login as Patient	Valid Patient credentials	Patient dashboard should appear.	Patient portal is implemented.	Pass	No issue found.
PAT_01	2. Open Medical History	Click Medical History	Medical history page should appear.	/patient/history route is implemented.	Pass	Page available.
PAT_01	3. View medical records	Patient record data	Records should be displayed chronologically.	Medical records are stored with record type, date, title and provider information.	Pass	Unified medical record structure implemented.
PAT_01	4. Review record details	Existing records	Patient should be able to review available medical information.	Patient portal contains medical history functionality.	Pass	No issue found.

4. Patient Prescription View

Tester Name: sinthia

Module Name: Admin – Prescriptions

Test Case ID: PAT_02

Test Scenario: A logged-in patient opens the prescription section.

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
PAT_02	1. Login as Patient	Valid credentials	Patient dashboard should load.	Patient portal is implemented.	Pass	No issue found.
PAT_02	2. Open Prescriptions	Click Pre-script ions	Prescription list should appear.	/patient/prescription route is implemented.	Pass	Page available.
PAT_02	3. Select a prescription	Existing prescription	Prescription details should be displayed.	Prescription records are supported in the database.	Pass	No issue found.
PAT_02	4. Review medication information	Diagnosis/medication	Medication details should be visible.	Prescription schema stores diagnosis, medications and instructions.	Pass	Prescription structure implemented.

5. Patient Diagnostic Reports

Tester Name: Mashuk

Module Name: Diagnostic Reports

Test Case ID: PAT₀3

Test Case Description: To check whether a patient can view their diagnostic test re-ports.

Test Scenario: Patient accesses diagnostic reports from the portal.

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
PAT_03	1. Login as Patient	Valid credentials	Patient dashboard should appear.	Patient portal is implemented.	Pass	No issue found.
PAT_03	2. Open Diagnostic Reports	Click Reports	Diagnostic report list should appear.	/patient/reports route is implemented.	Pass	Page available.
PAT_03	3. Select a report	Existing diagnostic report	Report details should be displayed.	Diagnostic report records are supported.	Pass	No issue found.
PAT_03	4. Review report	Test name, category, summary	Relevant diagnostic information should be displayed.	Report structure supports these fields.	Pass	No issue found.

6. Patient Hospital Records

Tester Name: Imran

Module Name: Hospital Records

Test Case ID: PAT₀4

Test Case Description: To check whether a patient can view hospital visit and admission records.

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
PAT_04	1. Login as Patient	Valid credentials	Patient dashboard should load.	Patient portal is implemented.	Pass	No issue found.
PAT_04	2. Open Hospital Records	Click Hospital Records	Hospital record page should appear.	/patient/hospital-records route is implemented.	Pass	Page available.
PAT_04	3. Select a hospital record	Existing visit record	Visit information should be displayed.	Hospital visit data is supported.	Pass	No issue found.
PAT_04	4. Review visit details	Visit type, department, date, reason	Relevant visit information should be available.	Database supports these fields.	Pass	Hospital record structure implemented.

Chapter 6

Conclusion

6.1 Discussion

The EHealth system addresses the major challenges of fragmented and paper-based healthcare records in Bangladesh. It provides a centralized platform for storing and accessing prescriptions, diagnostic reports, treatment histories, and appointment information. This improves patient record accessibility, reduces data loss and duplicate tests, and helps doctors make better treatment decisions. Doctor verification and prescription tracking also improve transparency and accountability. Overall, the system provides a secure and efficient foundation for improving digital healthcare services in Bangladesh.

6.2 Limitations

The limitations of the EHealth system are:

- Requires reliable internet connectivity for accessing online health records.
- Data security and patient privacy must be continuously maintained.
- The system depends on accurate information entered by healthcare providers.
- Some rural areas may face limited access to the required digital infrastructure.
- Integration with existing hospital and diagnostic center systems may be challenging.
- Users may require proper training to use the system effectively. Large-scale implementation may require significant financial and technical resources.

6.3 Scope of Future Work

The future scope of the EHealth system includes:

- Integration of Artificial Intelligence (AI) for disease prediction and clinical decision support.
- Implementation of blockchain technology to improve data security and record integrity.
- Integration with hospitals, diagnostic centers, pharmacies, and government healthcare services.
- Development of a mobile application for easier access to healthcare services.
- Enhancement of telemedicine features for remote consultation. Use of anonymized healthcare data for disease monitoring and healthcare planning.
- Implementation of stronger security and privacy mechanisms to protect sensitive patient information.

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- [1] Grameenphone, “GP Giving – Social and Humanitarian Donation Initiatives,” Grameenphone Bangladesh. [Online]. Available: <https://www.grameenphone.com/>
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